

**0:20 (cumulative 0:20)**

I am Takashi Yoshino, Toyo University. This talk is about nets of 4-dimensional polytopes. It closes three open cases.

A **net** of a convex polytope: cut it open and lay the facets out flat, without overlap. In 3D this is Dürer's problem, still open. In 4D the facets are **3-dimensional cells** and the net lives in  $\mathbb{R}^3$ .

- Turney (1984): all 261 nets of the 8-cell
- Devadoss–Harvey (2023): every spanning tree works for the 5-, 8-, 16-cell

Left open: 24-cell, 120-cell, 600-cell.



A net of the 8-cell: Dali's cross

slide 2

### 1:20 (cumulative 1:40)

Cut it open, lay it flat, no overlap. In 3D this is Dürer's problem. In 4D the pieces are solids, so the net is a 3D object you can print.

The picture: the 8-cell is the tesseract, and our net is Dali's cross — one of Turney's 261. It is forced, not luck. Two cells are adjacent unless they are opposite, so the layers are  $1 + 6 + 1$  and all 6 trees give this cross.

Devadoss–Harvey settled the three small ones for *every* spanning tree. End on the last line: three polytopes left open. That is where the talk starts.

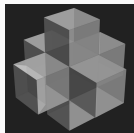
**Click the picture** to make it turn. Do not wait for it.

## Face-rotation BFS

Fix a **root cell**  $r$ . Three steps.

- 1 **Face-up.** Rotate in  $SO(4)$  so the centroid of  $r$  sits on the  $+w$  axis.
- 2 **BFS.** Take a breadth-first spanning tree of the cell-adjacency graph from  $r$ . For each tree edge (parent  $p$ , child  $c$ ), rotate  $c$  about the shared polygon until it lands in  $p$ 's hyperplane.
- 3 **Project.** All cells now share one  $w$ -coordinate (spread  $< 4 \times 10^{-13}$ ). Drop  $w$ .

The result is a net iff no two cells share interior volume.



Opening one BFS layer at a time, into Dali's cross

slide 3

## 1:20 (cumulative 3:00)

The slowest slide. The rotation is forced: the shared polygon is the hinge. Step 3 is a check, not an assumption — the cells really do land in one hyperplane. Do not dwell on step 2; the next slide is about what it leaves open.

**Click the picture** and let it run while you read the three steps. It opens *by layer*: the six neighbours first, then a pause, then the last cell. On that pause, say that BFS fixes the layers. If asked why the cells look bent: it is a 4D view, and it fades out as the net flattens. The last frame is the real net.

## Step 2 hides a choice

The layers are forced:  $L_d$  is the set of cells at distance  $d$  from  $r$ , and no rule can change that. The **parent** is not. A cell in  $L_d$  may touch several cells of  $L_{d-1}$ , and BFS takes whichever one the implementation happens to reach first.

### A geometric rule need not help

For the 5-, 8-, 16- and 24-cell the symmetries fixing  $r$  and  $v$  permute the candidates transitively, so **every** equivariant score — height along  $w$ , turning angle, the volume form — ties on all of them.

Ours breaks the tie by cell index, which is not equivariant either. **We come back to this.**



Two admissible parents  
for  $v$

slide 4

**0:50 (cumulative 3:50)**

Short, but this is the slide the referees care about. Two beats. One: BFS fixes the layers, not the parents. Two: a cleverer geometric rule does not help. For those four the candidates are interchangeable under the symmetry group, so any equivariant score ties on all of them.

Checked on every cell with more than one candidate: 1, 11 and 7 cells for the 8-, 16- and 24-cell, each a single orbit. The 5-cell has none.

**If pressed:** it fails for the 120-cell, where 50 of the 71 such cells have candidates in two or more orbits. Do not volunteer this.

Close with “we come back to this”.

### The validity test, and a trap

A net is valid if for every pair of cells not joined by a tree edge,

$$\text{Vol}_3(\text{conv}(c_i) \cap \text{conv}(c_j)) \leq 10^{-8}.$$

#### Boundary contact is not overlap

Testing with closed-region membership instead counts a shared face, edge or vertex as an intersection. That is common for pointed cells, and it wrongly reports

16-cell 0/16, 24-cell 0/24, 600-cell 0/600.

Measuring *volume* ignores contact of measure zero.

### 1:20 (cumulative 5:10)

Tell it as something that happened.

The first run said these three polytopes failed completely.

But cells that meet along an edge do not overlap. The fix is one line of code. Without it the theorem comes out backwards.

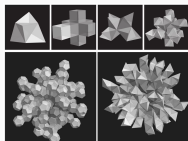
**Theorem**

For every regular convex 4-polytope and **every** root cell, the face-rotation BFS net is valid.

$5 + 8 + 16 + 24 + 120 + 600 = 773$  roots, all verified.

New beyond

Devadoss–Harvey: the 24-, 120- and 600-cell.



slide 6

**1:40 (cumulative 6:50)**

State the theorem, then let the picture sit. Six polytopes, one net each, all from root cell 1.

Say the number out loud: 773 roots, no exceptions. The overlap count is exactly zero every time — there are no near misses.

The three new ones are the three Devadoss–Harvey left open.

**Click the figure** and the two big nets turn, both at once. Eight seconds a turn. Let them run while you talk.

## Contrast: the apple-peel algorithm

A different family of spanning trees — a greedy spiral/zonal ordering (Yoshino–Chaidee, arXiv:2605.30373).

| Polytope | BFS       | apple-peel (RZ) |        |
|----------|-----------|-----------------|--------|
|          | valid     | orderings       | nets   |
| 120-cell | all roots | 1440/1440       | 0/1440 |
| 600-cell | all roots | 0/2400          | —      |

The 120-cell has an ordering for every start and **never** a net; the 600-cell has no ordering at all.

### Consequence

Having a valid *ordering* and having a valid *net* are independent properties.

## 1:30 (cumulative 8:20)

The companion algorithm peels the polytope like an apple, one cell at a time.

Read the two rows as opposite failures. The 120-cell always finds an order, and the order is never a net. The 600-cell never finds an order at all.

Then the block: ordering and net are independent. Do not explain how the apple-peel rule works. Only the contrast matters.

The arXiv number is there for anyone who wants the preprint. No need to read it out — just leave the slide up long enough to photograph.

## And more: the tie-breaking question

The abstract left this open:

*"Whether validity extends  
to all BFS tie-breakings ...  
remains open."*

BFS layers do not depend on the rule, so a BFS tree is just a choice of parent, one layer up, for each cell. They can then be counted with no rule at all.

**1,132,661 trees, no self-intersection.** For the first four the rule is irrelevant. And BFS is not a vacuous restriction: the 600-cell (Devadoss–Harvey) and the 24-cell (Seamons 2026) are **not all-net**.

|          | trees /                   | root      | checked |
|----------|---------------------------|-----------|---------|
| 5-cell   |                           | 1         | all     |
| 8-cell   |                           | 6         | all     |
| 16-cell  | 20,736                    |           | all     |
| 24-cell  | 32,768                    |           | all     |
| 120-cell | $\sim 1.8 \times 10^{44}$ | 20 random |         |
| 600-cell | $\sim 1.8 \times 10^{65}$ | 20 random |         |

slide 8

## 1:50 (cumulative 10:10)

Quote the abstract first. This answers the question the abstract asked; it is not a side remark.

The idea: the layers depend only on distance from the root, so a BFS tree is one parent per cell from the layer above, and the family can be counted without naming a rule. Four exhausted, two sampled; if asked why only 20 per root,  $10^{44}$ .

The last sentence is the payoff: it answers “why BFS and not every tree?”. Two of the six are not all-net, and BFS works on both. Say it as the reason, not an apology.

**If asked:** the 600-cell is their own Theorem 11; for the 24-cell, Seamons *confirms* their conjecture, in a 2026 preprint I checked myself.

⚠ Warning: this does not extend

Beyond the regular case the rule stops being harmless.

**Grand antiprism, 320 cells**

|        |                          |
|--------|--------------------------|
| rule A | valid from all 320 roots |
| rule B | valid from no root       |

Same polytope, same roots, same family of BFS trees. Only the choice of parent differs.

Among the 64 convex uniform 4-polytopes this is not isolated: another swings from 1.8% to 99.8% of its roots.

**What regularity buys is not that some tree works, but that the choice does not matter.**



Grand antiprism, one valid net

slide 9

## 1:40 (cumulative 11:50)

This follows from the last slide, so say “and this is exactly what fails once you leave the regular case”.

The grand antiprism is the cleanest case: all 320 roots or none, decided by nothing but which parent you pick.

Then the closing sentence. Say it slowly; it is the line to remember.

**If asked what A and B are:** A is the textbook queue rule — max-index does the same here — and B is “take the smallest-indexed candidate parent”. The indices carry no geometry. B2 has the numbers.

- Why? A proof that does not enumerate.
- From BFS trees to **all** spanning trees: the 600-cell is not all-net (Devadoss–Harvey), nor is the 24-cell — Seamons (2026) confirms their conjecture. **Only the 120-cell is left.**
- For the 120- and 600-cell even the BFS family is open:  $10^{44}$  and  $10^{66}$  trees per root put enumeration out of reach, so an argument would have to be structural.
- What separates the uniform polytopes that are insensitive to the rule from those that are not?

Code, data and nets:

[github.com/takashi-randomwalker/apple-peel-4d](https://github.com/takashi-randomwalker/apple-peel-4d)

slide 10

## 0:40 (cumulative 12:30)

Do not read all four. Say the second and the last, point at the URL, and stop — “only the 120-cell is left” is the line worth saying. If pressed: producing nets can never refute that conjecture; one overlapping tree confirms it, refuting it needs all of them. The slot ends at 16:57, leaving about two minutes for questions.

Backups: B1 reachable trees, B2 the three rules, B3 the runcinated 24-cell, B4 what does *not* separate them, B5 the data of the six.